Advanced Statistical Learning

Basis Expansions and Regularization

Xiaolei Lin

School of Data Science
Fudan University

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Outline

Piecewise Polynomials and Splines

Smoothing Splines

Nonparametric Logistic Regression

Multidimensional Splines

(Generalized) Linear Models

- Linear regression
- Linear discriminant analysis
- Logistic regression
- Separating hyperplanes

Extremely unlikely that $f(x)$ is linear

- Linear model is usually a convenient and sometimes necessary approximation
- Easy to interpret, first order Taylor expansion, avoid overfitting

A Taylor series expansion of a function $f(x)$ about a point a

- $f(x) =$
  $f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f^{(3)}(a)}{3!}(x-a)^3 + \ldots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \ldots$

Add-on

A Taylor series expansion of a vector function $f(x, y)$ about a point (a, b)

- $f(x, y) \approx f(a, b) + \frac{df}{dx}(a, b)(x - a) + \frac{df}{dy}(a, b)(y - b)$

approximation by the tangent plane at (a, b)

Moving Beyond Linearity

Augment inputs X with transformations, then use linear models in the new space

$$\text{The } m_{th} \text{ transformation, } h_m(X) : \mathbb{R}^p \to \mathbb{R}$$

The new approximated function: $f(X) = \sum_{m=1}^{M} \beta_m h_m(X)$, $m = 1, \ldots, M$

Once basis functions $h_m(X)$ have been determined, model estimation proceeds as in ordinary linear regression.

Some simple and widely used $h_m(X)$

- $h_m(X) = X_m$, the original input variables
- $h_m(X) = X_j^2$ or $X_j X_k$, $O(p^d)$ terms for degree-d polynomial)
- $h_m(X) = \log(X_j)$, $\sqrt{X_j}$, $||X||$, ...
- $h_m(X) = I(L_m \leqslant X_k \leqslant U_m)$, indicator for region of X_k

Controlling for Model Complexity

Restriction methods: limit the class of functions before-hand

- $f(x) = \sum_{j=1}^{p} f_j(X_j) = \sum_{j=1}^{p} \sum_{m=1}^{M_j} \beta_{jm} h_{jm}(X_j)$
- limit the number of basis functions M_j used for each component function f_j .

Selection methods: include basis that improve model fit significantly

- Variable selection
- Stagewise greedy approaches, CART, MARS and boosting

Regularization methods: use the entire dictionary but restrict the coefficients

- Ridge
- LASSO

Piecewise Polynomials & Splines

Piecewise Polynomials - Assume 1-dim

A piecewise polynomial function $f(X)$ is obtained by dividing the domain of X into contiguous intervals, and representing f by a separate polynomial in each interval.

Piecewise constant

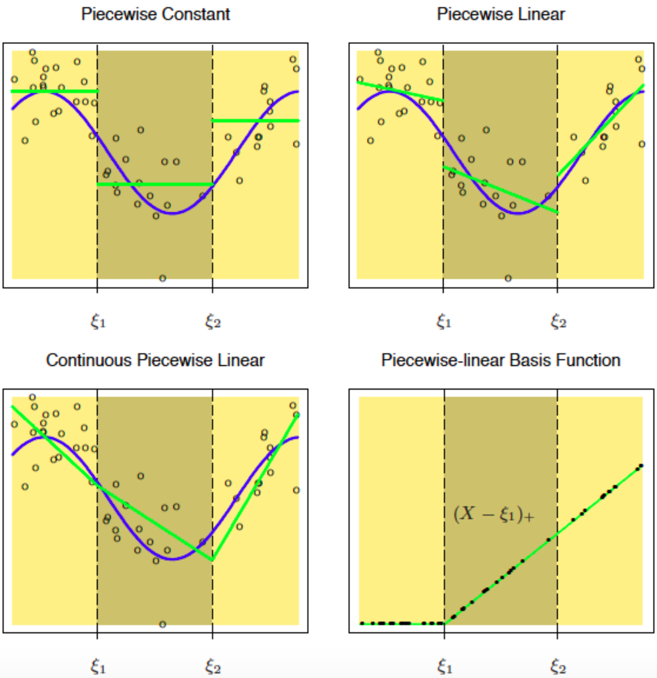
- $h_1(X) = I(X < \xi_1)$, $h_2(X) = I(\xi_1 \leqslant X < \xi_2)$, $h_3(X) = I(\xi_2 \leqslant X)$

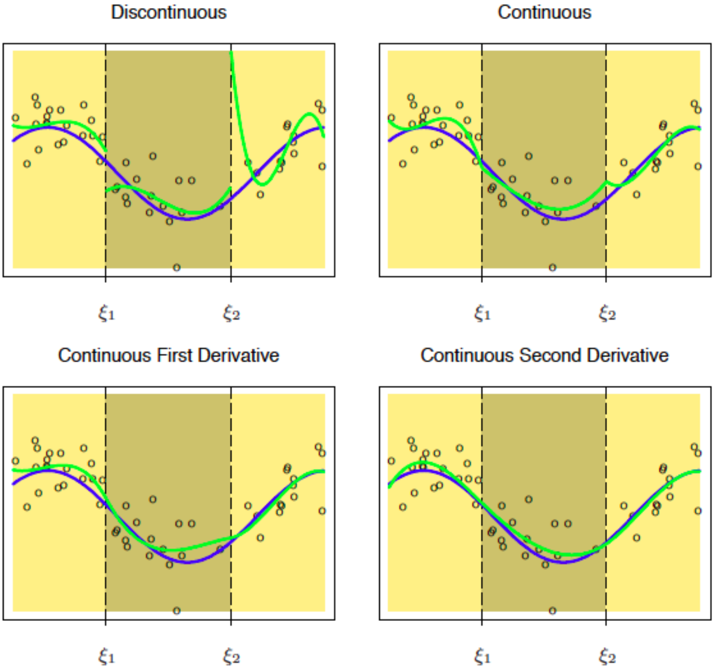
Piecewise linear

- $h_4(X) = I(X < \xi_1)X$, $h_5(X) = I(\xi_1 \leqslant X < \xi_2)X$, $h_6(X) = I(\xi_2 \leqslant X)X$

Continuous piecewise linear

- $h_1(X) = 1$, $h_2(X) = X$, $h_3(X) = (X - \xi_1)_+$, $h_4(X) = (X - \xi_2)_+$,


Piecewise Constant
Piecewise Linear
Continuous Piecewise Linear
Piecewise-linear Basis Function
ξ_1
ξ_2
$(X-\xi_1)_+$


Discontinuous
Continuous
Continuous First Derivative
Continuous Second Derivative
ξ_1
ξ_2

Piecewise Cubic Polynomial

Cubic spline: piecewise cubic polynomial with continuous first and second derivatives at the knots.

Basis for cubic splines

- $h_1(X) = 1$, $h_2(X) = X$, $h_3(X) = X^2$, $h_4(X) = X^3$
- $h_5(X) = (X - \xi_1)_+^3$, $h_6(X) = (X - \xi_2)_+^3$

Total number of parameters: $3 \times 4 - 2 \times 3 = 6$

Cubic splines are the lowest-order spline for which the knot-discontinuity is not visible to the human eye.

Order-M Spline

Order-M spline with knots ξ_j, $j = 1, ..., K$:

- Piecewise-polynomial of order M
- Continuous derivatives up to order M-2

Truncated power basis

- $h_j(X) = X^{j-1}$, $j = 1, ..., M$
- $h_{M+l} = (X - \xi_l)_+^{M-1}$, $l = 1, ..., K$

In practice, the most widely used orders are M $=$ 1, 2 and 4.

Construct Spline Basis

Choose order of the spline, the number of knots and their placement.

bs(x,df=7)

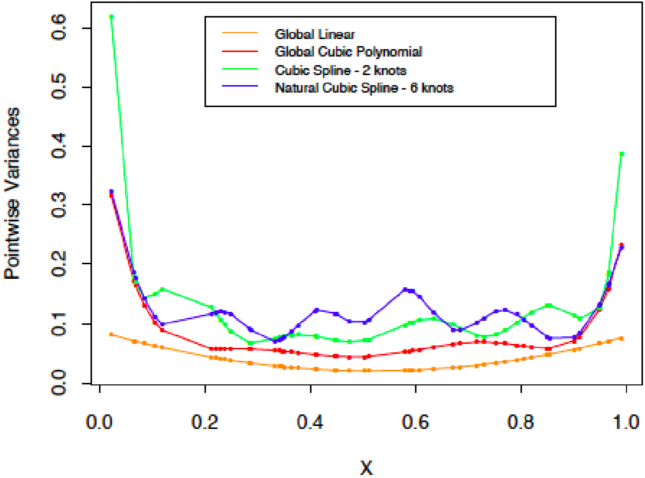
- Default order M=4 cubic spline
- 7-3=4 interior knots at the 20_{th}, 40_{th}, 60_{th}, 80_{th} percentiles of X
- Return a N $\times$ 8 matrix

bs(x, degree=1, knots = c(0.2, 0.4, 0.6))

- Piecewise constant
- return N $\times$ 4 matrix

Problems with polynomials and splines

- Poor fit near boundaries
- Extrapolation beyond boundaries



Natural Cubic Splines

Natural cubic spline

- Assume function is linear beyond the boundary knots
- Frees up 2×2 degrees of freedom

Natural cubic spline with K knots - K basis functions

- $N_1(X) = 1,\ N_2(X) = X,\ N_{k+2}(X) = d_k(X) - d_{K-1}(X)$
- $d_k(X) = \frac{(X-\xi_k)_+^3 - (X-\xi_K)_+^3}{\xi_K - \xi_k}$
- Easy to prove: zero second and third derivatives when $X \geqslant \xi_K$ or $X \leqslant \xi_1$

Example - South African Heart Disease

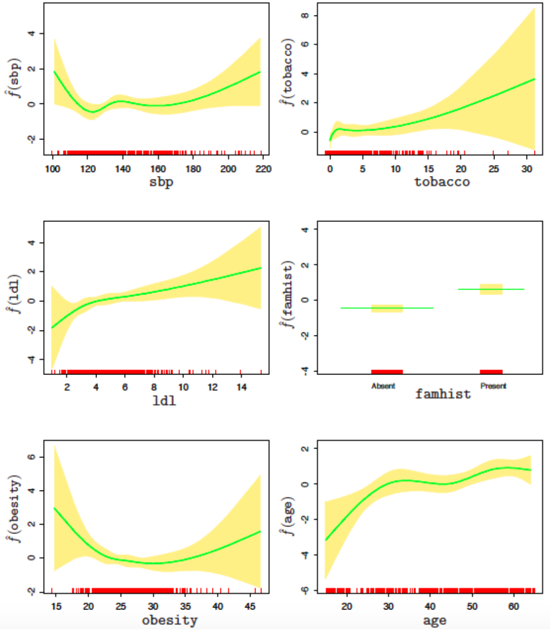
Modify the original logistic regression with

$$logitPr(chd \mid X) = \theta_0 + h_1(X_1)^\top \theta_1 + h_2(X_2)^\top \theta_2 + \ldots + h_p(X_p)^\top \theta_p$$

- four natural spline bases for each variable
- binary variables are dummy coded
- combine all p vectors of basis functions and estimate as a ordinary logistic regression

Terms	Df	Deviance	AIC	LRT	P-value
none		458.09	502.09		
sbp	4	467.16	503.16	9.076	0.059
tobacco	4	470.48	506.48	12.387	0.015
ldl	4	472.39	508.39	14.307	0.006
famhist	1	479.44	521.44	21.356	0.000
obesity	4	466.24	502.24	8.147	0.086
age	4	481.86	517.86	23.768	0.000

Example - South African Heart Disease



Smoothing Splines

Smoothing Splines

With regression splines, choosing knots is a tricky business

Smoothing Splines

- avoid knot selection problem by using a maximal set of knots
- control for overfitting by shrinking the coefficients of the estimated function

Smoothing splines can be motivated directly by 1): natural cubic splines with knots at $(x_{(1)}, x_{(2)}, ..., x_{(n)})$ and regularization on parameters, or 2): a function minimization perspective. We'll talk about 2) in accordance with the text.

Smoothing Splines

Among all functions $f(x)$ with two continuous derivatives, find one that minimizes the penalized residual sum of squares

$$RSS(f, \lambda) = \sum_{i=1}^{N} (y_i - f(x_i))^2 + \lambda \int f^{''}(t)^2 d(t)$$

- the first term quantifies the goodness-of-fit to the data
- the second term measures the roughness (curvature in f)
- λ: smoothing parameter

Special case

- $\lambda = 0$: f can be any function that interpolates the data.
- $\lambda = \infty$: f is simple least squares line

With $\lambda \in (0, \infty)$, there exists a unique minimizer - a natural cubic spline with knots at the unique values of the $x_i, i = 1, \ldots, N$

Proof

THEOREM :

Let g be any differentiable function on $[a, b]$ for which $g(x_i) = z_i$ for $i = 1, \dots, n$. Suppose $n \geqslant 2$, and that $\tilde{g}$ is the natural cubic spline interpolant to the values $z_1, \dots, z_n$ at points $x_1, \dots, x_n$ with $a < x_1 < \quad < x_n < b$.

Then $\int (g^{''})^2 \geqslant \int (\tilde{g}^{''})^2$ with equality only if $\tilde{g} = g$

hint: construct $h(x) = g(x) - \tilde{g}(x)$ such that $h(x_i) = 0$ for $i = 1, \dots, n$. Then show that $\int [\tilde{g}''(x)]^2 dx \leq \int [g(x)]^2 dx$

Smoothing Splines

Since now $f(x)$ is a natural cubic spline, $f(x) = \sum_{j=1}^{N} N_j(x)\theta_j$

The penalized RSS reduces to:

$RSS(\theta, \lambda) = (y - N\theta)^\top (y - N\theta) + \lambda\theta^\top \Omega_N \theta$

- N is $n \times n$, with $N_{ij} = N_j(X_i)$
- Ω_N is $n \times n$, with $\Omega_{N\,ij} = \int N_i''(t) N_j''(t) dt$

The solution: $\hat{\theta} = (N^\top N + \lambda\Omega_N)^{-1} N^\top y$

The fitted: $\hat{y} = N\hat{\theta} = N(N^\top N + \lambda\Omega_N)^{-1} N^\top y = S_\lambda y$

S_λ: smoother matrix

Pre-specify the Amount of Smoothing

Assume λ is fixed, smoothing spline operates through linear operator

$\hat{f} = N(N^\top N + \lambda\Omega_N)^{-1}N^\top y = S_\lambda y$

- fits are linear in y
- smoother matrix S_λ depends on X and λ

Compare hat matrix H and smoother matrix S_λ

- symmetric, positive semidefinite?
- idempotent?
- rank?

Effective degree of freedom $df_\lambda = tr(S_\lambda)$

Smoother Matrix

1. Re-write S_λ in the Reinsch form

$$S_\lambda = (I + \lambda K)^{-1}$$

- $K = N^{-\top}\Omega_N N^{-1}$, does not depend on λ
- K: penalty matrix since $PRSS = (y - f)^\top (y - f) + \lambda f^\top K f$

2. Eigen decomposition of S_λ

$$S_\lambda = \sum_{k=1}^{N} \rho_k(\lambda) \mu_k \mu_k^\top$$

- Eigenvalue $\rho_k(\lambda) = \frac{1}{1+\lambda d_k}$
- Eigenvector μ_k

Note: μ_k are also the eigenvector of K, and d_k are the eigenvalue of K

Highlights of Eigen Representation

$$S_\lambda = \sum_{k=1}^{N} \rho_k(\lambda)\mu_k\mu_k^\top$$

- The eigenvectors are not affected by λ
- $S_\lambda y = \sum_{k=1}^{N} \mu_k \rho_k(\lambda) < \mu_k^\top y >$: decompose y with respect to new basis μ_k, then shrink it using $\rho_k(\lambda)$.
- compare to regression spline with M basis?
  - projection matrix H: M eigenvalues=1, N-M eigenvalues=0, components either left alone, or shrink to 0
  - smoother matrix S_λ: shrinking smoother
- $d_1 = d_2 = 0$, thus $\rho_1(\lambda) = \rho_2(\lambda) = 1$, linear functions are never shrunk

Highlights of Eigen Representation

$$S_\lambda = \sum_{k=1}^{N} \rho_k(\lambda) \mu_k \mu_k^\top$$

- The eigenvectors have increasing complexity, and the higher the complexity, the more they are shrunk
- Reparametrize RSS with new basis μ_k, $RSS = ||y - U\theta|| + \lambda\theta^\top D\theta$, thoughts? (ridge regression on the new basis)
- $df_\lambda = tr(S_\lambda) = \sum_{k=1}^{N} \rho_k(\lambda)$, compare to regression splines?

Add-on

Assume N is invertible (can be proved, det is non-zero)

$$
\begin{aligned}
S_\lambda &= N(N'N + \lambda\Omega_N)^{-1}N' \\
&= N(N'N + \lambda N'(N')^{-1}\Omega_N N^{-1}N)^{-1}N' \\
&= N(N'(I + \lambda(N')^{-1}\Omega_N N^{-1})N)^{-1}N' \\
&= NN^{-1}(I + \lambda(N')^{-1}\Omega_N N^{-1})^{-1}(N')^{-1}N' \\
&= (I + \lambda(N')^{-1}\Omega_N N^{-1})^{-1} \\
&= (I + \lambda K)^{-1}
\end{aligned}
$$

where $K = (N')^{-1}\Omega_N N^{-1}$

Add-on

$K = (N')^{-1}\Omega_N N^{-1}$ symmetric and positive semi-definite, eigen decomposition: $K = PDP^{-1}$

$$\begin{aligned}
S_\lambda &= (I + \lambda K)^{-1} \\
&= (I + \lambda PDP^{-1})^{-1} \\
&= (PP^{-1} + \lambda PDP^{-1})^{-1} \\
&= (P(I + \lambda D)P^{-1})^{-1} \\
&= P(I + \lambda D)^{-1}P^{-1} \\
&= \sum_{k=1}^{n} \frac{1}{1 + \lambda d_k} \mu_k \mu_k'
\end{aligned}$$

where d_k is eigenvalue of K, and μ_k is the corresponding eigen vector. Therefore, eigenvalue for S_λ is $\frac{1}{1+\lambda d_k}$

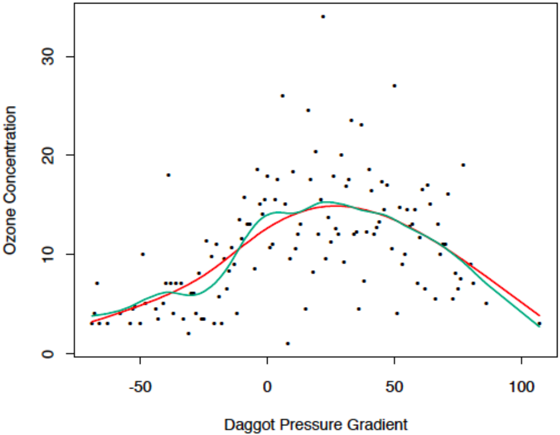
Add-on

Show that $PRSS = (y - f)'(y - f) + \lambda f' K f$

$$
\begin{aligned}
PRSS &= (y - f)'(y - f) + \lambda \theta' \Omega_N \theta \\
&= (y - f)'(y - f) + \lambda (N^{-1} N \theta)' \Omega_N (N^{-1} N \theta) \\
&= (y - f)'(y - f) + \lambda \theta' N' (N')^{-1} \Omega_N N^{-1} N \theta \\
&= (y - f)'(y - f) + \lambda f' K f
\end{aligned}
$$

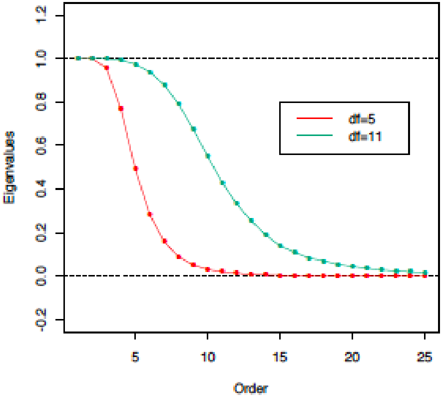
where $K = (N')^{-1} \Omega_N N^{-1}$

Example: Smoothing spline to air pollution data



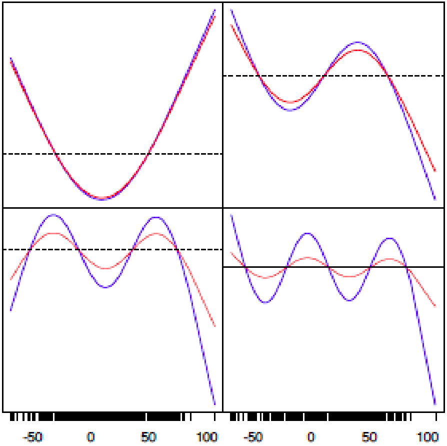
- Green curve: smoothing spline with $df_\lambda = tr(S_\lambda) = 11$
- Red curve: smoothing spline with $df_\lambda = tr(S_\lambda) = 5$

Example: Eigenvalues of S_λ



- Green curve: eigenvalues of S_λ with $df_\lambda = 11$
- Red curve: eigen values of S_λ with $df_\lambda = 5$

Example: Eigenvector of S_λ



- Each blue curve is an eigenvector of S_λ plotted against x. Corresponding eigenvalue decrease from top left to bottom right.
- Red curve: eigenvector damped by $\frac{1}{1+\lambda d_k}$

Choosing λ?

Crucial and tricky...

1. Fixing the Degrees of Freedom
   - $df_\lambda = tr(S_\lambda)$ is monotone in λ for smoothing splines
   - smooth.spline(x,y,df=6)
2. Cross validation
   - $EPE(\hat{f}_\lambda) = \sigma^2 + Var(\hat{f}_\lambda) + Bias(\hat{f}(\lambda))^2$
   - K-fold cross-validation
   - seek a good compromise between bias and variance

Nonparametric Logistic Regression

Logistic Regression

Logistic regression with single quantitative input X

$$\log \frac{Pr(Y=1|X=x)}{Pr(Y=0|X=x)} = f(x)$$

- Fitting f(x) in a smooth fashion leads to a smooth estimate of the conditional probability $Pr(Y = 1 \mid x)$

The likelihood function, where $Pr(Y = 1 \mid X = x) = p(x_i)$

$$l = \sum_{i=1}^{N} \{y_i \log p(x_i) + (1 - y_i) \log(1 - p(x_i))\}$$

since $p(x_i) = \frac{e^{f(x_i)}}{1+e^{f(x_i)}}$,

$$l = \sum_{i=1}^{N} \left\{y_i f(x_i) - \log(1 + e^{f(x_i)})\right\}$$

Penalized log-likelihood

$$l = \sum_{i=1}^{N} \left\{ y_i f(x_i) - \log(1 + e^{f(x_i)}) - \frac{1}{2}\lambda \int \left\{ f''(t) \right\}^2 dt \right\}$$

Optimal f is a finite-dimensional natural spline with knots at the unique values of x.

Represent $f(x) = \sum_{j=1}^{N} N_j(x)\theta_j$

- $\frac{\partial l(\theta)}{\partial \theta} = N^\top (y - p) - \lambda \Omega \theta$
- $\frac{\partial^2 l(\theta)}{\partial \theta \partial \theta^\top} = -N^\top W N - \lambda \Omega$

Newton-Raphson update

- $\theta^{new} = (N^\top W N + \lambda \Omega)^{-1} N^\top W (N\theta^{old} + W^{-1}(y - p)) =$ $(N^\top W N + \lambda \Omega)^{-1} N^\top W * z$
- $f^{new} = N(N^\top W N + \lambda \Omega)^{-1} N^\top W (f^{old} + W^{-1}(y - p)) = S_{\lambda, W} z$

Multidimensional Splines

From 1-dim to 2-dim

Suppose $X \in \mathbb{R}^2$

- $h_{1j}(X_1)$, $j = 1, \dots, M_1$ are the basis for coordinate X_1
- $h_{2k}(X_2)$, $k = 1, \dots, M_2$ are the basis for coordinate X_2
- $g_{jk}(X) = h_{1j}(X_1)h_{2k}(X_2)$ are the $M_1 \times M_2$-dim tensor product basis

The new basis representation of the model

$$g(X) = \sum_{j=1}^{M_1} \sum_{k=1}^{M_2} \theta_{jk} g_{jk}(X)$$

Note, the dimension of the basis grows exponentially fast - curse of dimensionality.

From 1-dim to 2-dim

The 1-dim penalty

$$\lambda \int f^{''}(t)^2 dt$$

Natural generalization to 2-dim

$$\int \int_{\mathbb{R}^2} \left\{ \left(\frac{\partial^2 f(x)}{\partial x_1^2}\right)^2 + 2\left(\frac{\partial^2 f(x)}{\partial x_1 \partial x_2}\right)^2 + \left(\frac{\partial f(x)}{\partial x_2^2}\right)^2 \right\} dx_1 dx_2$$

The solution is a smooth two-dimensional surface - thin plate spline

- $\lambda \to 0$, $f \to$ interpolating function
- $\lambda \to \infty$, $f \to$ least square plane
- For $\lambda \in (0, \infty)$, solution has the form $f(x) = \beta_0 + \beta^\top x + \sum_{j=1}^{N} \alpha_j h_j(x)$
- $h_j(x) = \|x - x_j\|^2 \log \|x - x_j\|$, radial basis function

d-dim

More generally, when $X \in \mathbb{R}^d$

$$\min \sum_{i=1}^{N} \{y_i - f(x_i)\}^2 + \lambda J \mid f \mid$$

Restricted class of multidimensional splines

$$J \mid f \mid = J(f_1 + f_2 + \ldots + f_d) = \sum_{j=1}^{d} \int f_j^{''}(t_j) dt_j$$

- f is additive
- additive penalty on each of the component functions